

Serial No. 056158

Sub Code: 23UCPGE3

H.H.THE RAJAH'S COLLEGE (AUTONOMOUS B ), PUDUKKOTTAI -622 001

B.Sc. DEGREE EXAMINATIONS – APRIL - 2026

PART III - COMPUTER SCIENCE – GENERIC ELECTIVE COURSE – III  
ALLIED PHYSICS PAPER

Time: 3 hours

Max: 75 marks  
(10x2=20marks)

**Section – A**

**Answer ALL the Questions**

1. Define electric field.
2. Define magnetic susceptibility.
3. State Kirchhoff's law.
4. What is eddy current?
5. What is semiconductor?
6. Define transistor.
7. State the properties of laser.
8. What is population inversion?
9. What is an amplifier?
10. Give an acronym of LED & LCD.

**Section – B**

(5x5=25marks)

**Answer ALL the Questions**

11. a) Explain the principle and working of a capacitor.  
(OR)  
b) Derive an expression for the potential due to an electric dipole.
12. a) Explain the principle and working of a potentiometer.  
(OR)  
b) Explain AC circuit containing resistance only.
13. a) Distinguish between intrinsic and extrinsic semiconductor.  
(OR)  
b) Write a short note on Zener diode and its application.

14. a) Describe the working of Ruby laser.

(OR)

b) Write short notes on population inversion and pumping action.

15. a) Explain inverting and non-inverting operational amplifier.

(OR)

b) What is photodiode? Explain its working.

**Section – C**

**(3x10=30marks)**

**Answer any THREE Questions Only**

16. Discuss the properties of dia, para and ferro magnetic substance with exam

17. Describe the construction, principle and working of a transformer.

18. Explain the working of n-channel FET and its characteristics.

19. Explain the construction and working of He-Ne laser.

20. Write short notes on (i) Seven segment display (ii) LCD construction and working.

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